

Topic 30 -- Separable Equations

Definition of Separable

A separable equation is a first-order differential equation in which the expression for $\frac{dy}{dx}$ can be factored as a function of x times a function of y .

$$\frac{dy}{dx} = f(x)g(y)$$

Example 1

Solve: $\frac{dy}{dx} = \frac{e^{2x}}{4y^3}$

Example 2

Solve: $\frac{dy}{dt} = \frac{te^t}{y\sqrt{1+y^2}}$

Example 3

Solve: $\frac{dv}{ds} = \frac{s+1}{sv+s}$

Example 4

Solve: $y' = \frac{8x^7y}{\ln y}$

Example 5

Solve: $\frac{dy}{dx} = \frac{1+x}{xy}$; $x > 0$; $y(1) = -4$

Modeling with Separable Equations

Example 6

“A population grows at a rate proportional to its size.”

Example 7

A tank contains 50 gallons of solution at a concentration of 2lbs/gal. Pure water is pumped in at the rate of 4 gal/min. The mixture is well stirred and drained at the same rate. Find the amount of salt, in pounds, in the tank at any time, measured in minutes.

Example 8

A tank initially holds 10 gal of fresh water. At $t = 0$, a brine solution containing 0.5 lb of salt per gallon is poured into the tank at a rate of 2 gal/min, while the well-stirred mixture leaves the tank at the same rate. Find the amount of salt, in pounds, in the tank at any time, measured in minutes.